Robustness Benchmarking of Embedded Battery State Estimation: A Measurement-Only Comparison of Direct-Measurement, Hybrid, and Data-Driven Estimators

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Abstract

This paper presents a measurement-only robustness benchmark for embedded battery state estimation under realistic Battery Management System (BMS) constraints. Four embedded-ready estimators are compared on the same Lithium Iron Phosphate (LFP) full-cell trajectory: a direct-measurement model (DM), a hybrid direct-measurement model with learned State-of-Health support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven neural model with joint SOC/SOH inference (DD). The benchmark perturbs only measurable inputs and timing information, including current bias, current noise, missing samples, irregular sampling, burst dropouts, spikes, and initial-state mismatch, so that all disturbances remain physically interpretable and reproducible without introducing artificial model-parameter mismatch. On the benchmark cell, HDM provides the best clean-condition baseline with an MAE of 0.0042, followed by HECM with 0.0087, DD with 0.0191, and DM with 0.0311. Under realistic current bias, HECM is clearly the most robust estimator, reaching an MAE of 0.0984 at a +3% current bias, compared with 0.2843 for DD, 0.3102 for HDM, and 0.3199 for DM. Under missing samples, DD remains particularly stable, changing only from 0.0191 to 0.0199 MAE, whereas the integration-based estimators degrade much more strongly. A CV-free local recovery analysis for initial-state perturbations shows that DM, HECM, and DD recover within about 1.16–1.18 h under a strict baseline-band criterion,

whereas HDM does not recover within the inspected window. The benchmark therefore exposes complementary failure modes of direct-measurement, hybrid physics-based, and data-driven estimators and supports decision-oriented model selection for embedded BMS deployment.

Keywords: Battery management system, State of charge, State of health, Robustness benchmark, Direct measurement, Equivalent circuit model, Data-driven estimation, Embedded artificial intelligence

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Nomenclature

Acronyms

BMS	Battery management system
SOC	State of charge
SOH	State of health
DM	Direct-measurement model
HDM	Hybrid direct-measurement model
HECM	Hybrid equivalent-circuit model
DD	Data-driven neural model
ECM	Equivalent circuit model
EKF	Extended Kalman filter
MAE	Mean absolute error
RMSE	Root-mean-square error
ADC	Analog-to-digital converter
CV	Constant-voltage phase
OCV	Open-circuit voltage
GRU	Gated recurrent unit
LFP	Lithium iron phosphate

Symbols

t	Time
$I(t)$	Current signal at time t
$U(t)$	Voltage signal at time t
$T(t)$	Temperature signal at time t
$\widehat{\mathrm{SOC}}(t)$	Estimated state of charge at time t
$\widehat{\mathrm{SOH}}(t)$	Estimated state of health at time t
$\mathrm{SOC}_{\mathrm{ref}}(t)$	Reference SOC target at time t
$\mathrm{SOH}_{\mathrm{ref}}(t)$	Reference SOH target at time t
ΔI	Current bias or offset
ΔU	Voltage bias or offset
ΔT	Temperature bias or offset
σ_I	Standard deviation of injected current noise
σ_U	Standard deviation of injected voltage noise
σ_T	Standard deviation of injected temperature noise
Q_c	Online cumulative charge proxy used by the SOC models
C_{nom}	Nominal capacity
C_{eff}	Effective capacity used inside the estimator
ΔMAE	Error increase relative to baseline

1. Introduction

- State estimation in BMS applications is usually reported through offline accuracy, but embedded deployment quality depends equally on robustness under realistic signal corruption.

- The core engineering problem is not only how well a model predicts under clean measurements, but whether it drifts, becomes unstable, or fails to recover when current, voltage, temperature, or timing information are corrupted.

- Existing comparisons often mix fundamentally different uncertainty sources: dataset shift, model mismatch, synthetic parameter errors, and sensor disturbances. This work narrows the scope deliberately to *measurement-only robustness*.

- The paper compares four estimators that represent different state-estimation philosophies while sharing the same measurement stream and reference targets.

- The intended contribution is a decision-oriented robustness benchmark that answers: when does each estimator fail, why does it fail, how large is the error increase, and does it recover without unrealistic recalibration assumptions.

- The benchmark is designed around reproducibility: same cell, same reference trajectory, same manipulated measurements, same output metrics, and consistent online execution assumptions.

2. Related Work and Motivation

2.1. SOC/SOH estimation families

- structure the literature into direct-measurement, model-based or physics-based, hybrid, and purely data-driven approaches.

- summarize the known advantages of direct-measurement approaches, especially simplicity, low compute cost, and intuitive online integration.

- summarize the known limitations of direct-measurement approaches, especially drift under current bias, missing samples, and sensitivity to initial SOC.

- summarize hybrid equivalent-circuit or ECM/EKF approaches, emphasizing voltage feedback, observability, and the potential to correct drift when informative voltage dynamics exist.

- summarize data-driven SOC and SOH approaches, especially sequence models that can absorb nonlinear relations but may be sensitive to corrupted inputs or out-of-distribution signal characteristics.
- position the present study as a robustness-focused comparison instead of a new estimator paper.

2.2. Gap in the literature

- most published battery-AI papers emphasize clean-data accuracy, not failure modes under realistic sensor faults.
- many studies test on new datasets or shifted temperatures, but fewer isolate strictly manipulable measurement disturbances.
- embedded papers often report runtime and memory but do not couple those measurements with robustness under sensor corruption.
- papers rarely compare integration-based, physics-based, and data-driven models inside a single measurement-only benchmark framework.
- this gap motivates a benchmark that is useful for method selection under BMS engineering constraints, not only for leaderboard accuracy.

2.3. Study objectives

- define robustness as a joint property of error increase, stability, drift, and recovery.
- compare one representative direct-measurement model, one hybrid direct-measurement model, one hybrid equivalent-circuit model, and one data-driven neural model under controlled measurement disturbances.
- map these paper-level classes to the implementation identifiers `CC\_1.0.0`, `CC\_SOH\_1.0.0`, `ECM\_0.0.1`, and `SOC\_SOH\_1.6.0.0\_GRU\_0.3.1.2`.
- identify which failure modes are structural, such as current-bias drift for direct-measurement models or spike sensitivity for data-driven models.

- translate the results into a model-selection guideline for embedded BMS deployment.

3. Dataset, Reference Targets, and Online Execution Assumptions

3.1. Cell data and benchmark scope

- describe the LFP full-cell measurement source and clarify that all robustness tests are executed on the same cell trajectory for the current benchmark phase, starting with `MGFarm\_18650\_C07`.
- describe available raw signals: current, voltage, temperature, test time, and reference SOC / SOH annotations.
- explain why the study deliberately avoids new datasets and synthetic cell parameters in this phase.
- explain that this keeps the benchmark controlled and isolates measurement corruption rather than domain shift.

3.2. Reference SOC and SOH

- explain that the reference SOC is derived offline using dataset-side logic that is not available in deployment.
- explain that the reference SOH is also an offline target and serves only as evaluation reference, not as an online input at runtime unless specifically estimated.
- clarify that the estimators operate online only on current, voltage, temperature, timing, and their own internal states.
- explicitly state that during disturbed scenarios the online auxiliary variables, such as cumulative charge proxies, must be recomputed from the disturbed measurements rather than reused from the clean dataset.

Placeholder for the overall benchmark pipeline: raw cell measurements → scenario manipulation → online estimator execution → global/local metrics → paper result package.
`figures/Schematics/benchmark_pipeline_overview.pdf`

Figure 1: Planned overview figure for the robustness benchmark pipeline.

3.3. Online execution principle

- explain that every estimator is executed as if it were deployed online.
- explain that missing or frozen measurements must also freeze or distort derived online features accordingly.
- explain that no hidden "oracle" variables from the dataset are allowed to leak into the disturbed runs.
- clarify how CV resets are handled in the direct-measurement estimators and why that matters for recovery interpretation.
- clarify how the hybrid equivalent-circuit estimator uses voltage feedback and how the data-driven estimator uses rolling sequential inference.

4. Estimators Under Test

4.1. Direct-measurement model (DM, implementation: `CC_1.0.0`)

- explain that the model integrates current into an online charge proxy and converts it into SOC using a fixed nominal capacity.
- explain the role of CV-triggered reset events.
- state the expected structural weakness under current bias and missing-sample conditions.
- note the expected strength under low-compute embedded deployment.

4.2. Hybrid direct-measurement model (HDM, implementation: `CC_SOH_1.0.0`)

- explain that the direct-measurement core remains integration-based, but effective capacity is adjusted through the GRU-based SOH estimator.
- describe how SOH is updated at a slower cadence and then held piecewise constant for online SOC estimation.
- emphasize that this should improve baseline calibration but not eliminate current-bias drift.
- flag that recovery from a wrong initial SOC may differ from the pure direct-measurement model because the capacity adaptation changes the mapping between integrated charge and SOC.

4.3. Hybrid equivalent-circuit model (HECM, implementation: `ECM_0.0.1`)

- describe the equivalent-circuit structure and the EKF-style state update in words.
- clarify that the model is a hybrid physics-based estimator because it integrates current but corrects using voltage residuals.
- describe how SOH-dependent capacity and parameter adaptation enter the equivalent-circuit implementation.
- motivate why the hybrid equivalent-circuit model should be more robust to current bias and initial SOC error than pure direct measurement.

4.4. Data-driven neural model (DD, implementation: `SOC_SOH_1.6.0.0_GRU_0.3.1.2`)

- describe the GRU-based SOH model and the SOC model that consumes voltage, current, temperature, SOH, and online auxiliary features.
- explain that the SOC model still depends on online features such as Q_c, so correct disturbance propagation matters.
- explain the implemented proxy for initial SOC error via initial perturbation of the online Q_c state.

- motivate why this model may handle some missing-sample scenarios well while still being sensitive to certain input spikes or bias patterns.

5. Robustness Benchmark Methodology

5.1. Principle: measurement-only disturbance design

- explain that only measurable inputs and timing are manipulated: current, voltage, temperature, sample availability, and sample spacing.
- explain why artificial ECM parameter mismatch and synthetic new SOH regimes are excluded from this benchmark phase.
- state that this design keeps the benchmark experimentally defensible and easy to reproduce.

5.2. Scenario classes

- define **input disturbances**: current noise, voltage noise, temperature noise, current offset, voltage offset, temperature offset, ADC quantization, and spikes.
- define **initialization errors**: initial SOC mismatch where structurally applicable.
- define **signal integrity problems**: missing samples, irregular sampling, and burst gaps.
- explain which scenarios are expected to be most informative for each model family.

5.3. Implemented scenario matrix

- list the compact benchmark phase already implemented: baseline, current noise, current offset, initial SOC error, missing samples, spikes.
- list the extended matrix that is now prepared for execution: voltage noise, temperature noise, voltage offset, temperature offset, ADC quantization, irregular sampling, missing gap, stronger current noise, stronger spikes.

- explain that scenario aliases are kept distinct in the campaign runner so that the paper tables can separate low and high noise as well as baseline and strong-spike cases.
- reference the benchmark runner files in the simulation environment.

5.4. Metrics

- define global error metrics: MAE, RMSE, bias, maximum error, and 95th percentile error.
- define local stability metrics: jump count above a fixed SOC threshold, output variance, and absolute-error variance.
- define disturbance-window metrics: pre-disturbance MAE, disturbed MAE, post-disturbance MAE, residual error after recovery.
- define recovery metrics: time to re-enter the baseline band and sustain the recovery.
- define drift metrics: late-minus-early rolling MAE and error slope over time.
- explain the difference between a strict baseline band and a fair baseline band for local recovery interpretation.

5.5. Implementation details and reproducibility

- identify the benchmark scripts and result folders in `4\_simulation\_environment`.
- explain how campaigns, manifests, and result tables are organized.
- describe how the `results/` workspace will collect only the paper-facing outputs.

Placeholder for a scenario taxonomy figure grouping input disturbances, initialization errors, and signal-integrity faults.

`figures/Schematics/scenario_taxonomy.pdf`

Figure 2: Planned taxonomy figure of the robustness scenarios used in the benchmark.

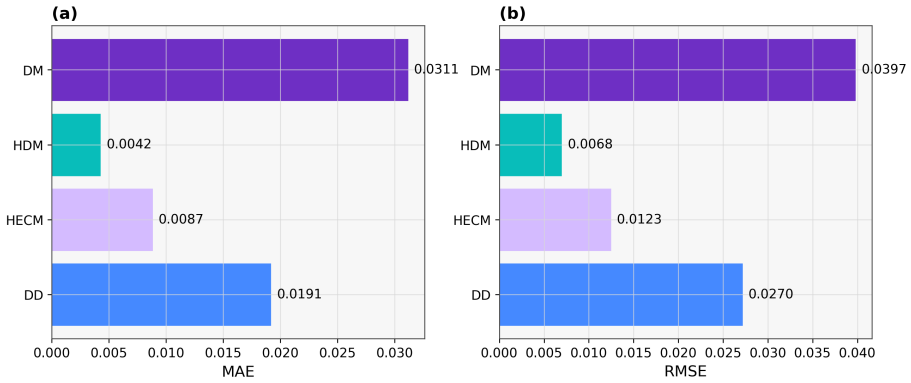

Figure 3: Baseline performance on the clean C07 full-cell trajectory. The hybrid direct-measurement model provides the lowest clean-condition error, whereas the direct-measurement model shows the largest baseline deviation.

6. Results

6.1. Baseline comparison

- On the clean C07 trajectory, the hybrid direct-measurement model achieves the lowest baseline error with MAE = 0.0042, followed by the hybrid equivalent-circuit model with 0.0087, the data-driven model with 0.0191, and the direct-measurement model with 0.0311.
- The baseline ranking therefore favors HDM for nominal operation, but this ordering does not persist under biased or corrupted measurements.
- This separation between clean-condition accuracy and robustness is one of the central findings of the benchmark.

6.2. Current bias / offset robustness

- Current bias is the clearest discriminator of integration-driven drift in the benchmark.
- Across the matched bias sweep (0.5%, 1.5%, and 3.0% of $|I|_{\max}$), the hybrid equivalent-circuit model remains the most robust estimator, the data-driven model is intermediate, and both direct-measurement models degrade strongly.
- The result is structurally consistent with the estimator classes: HECM can correct through voltage feedback, whereas DM and HDM remain dominated by current integration drift.
- The HDM keeps the best clean-condition baseline but does not gain practical protection against current bias once the current stream is systematically shifted.

6.3. Noise robustness

- Current noise is weaker than current bias in global MAE, but it remains informative through local behavior and output jitter.
- Under the high-noise case, the hybrid equivalent-circuit model shows the largest global penalty (ΔMAE = 0.0081), whereas DM, HDM, and DD remain close to their baseline MAE values.
- The local noise story differs from the global metric: the data-driven model exhibits the strongest output jitter, while the hybrid equivalent-circuit model is the most sensitive in the error domain.
- The figure therefore separates the applied input disturbance, a matched SOC comparison in the same window, the global MAE penalty, and the local output-jitter summary.

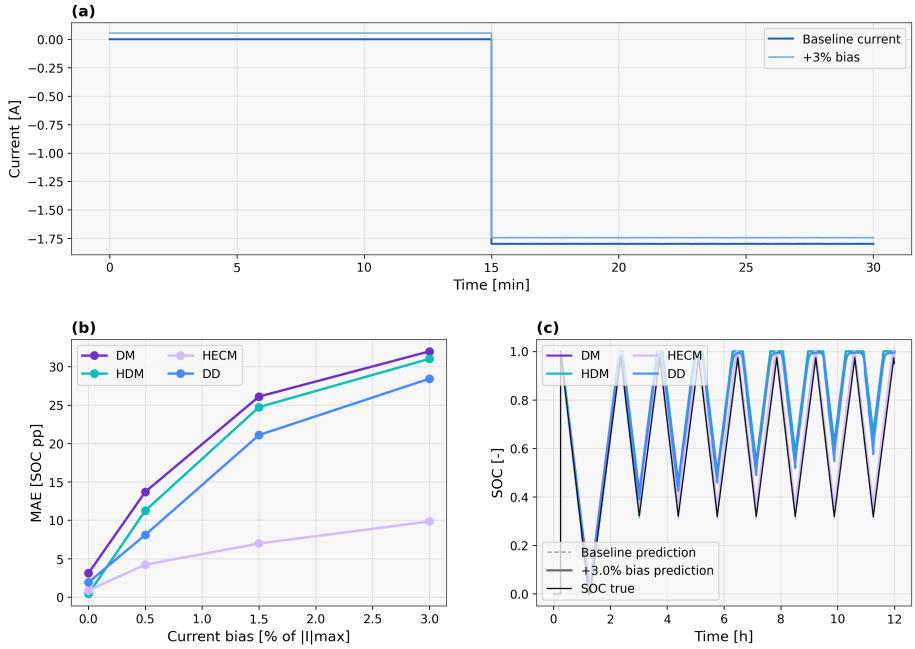

Figure 4: Robustness against current measurement bias. (a) Example of the applied current bias (+3%) in a representative 30 min window. (b) Global degradation of SOC estimation accuracy as a function of bias magnitude. (c) SOC trajectory divergence over the first 12 h for a +3% current bias.

6.4. Initialization error and recovery dynamics

- Initialization robustness must be analyzed locally and in a CV-free window, because global MAE hides the actual re-synchronization behavior after a wrong initial SOC.

- The direct-measurement and hybrid equivalent-circuit models return toward the reference trajectory, whereas the hybrid direct-measurement model shows persistent mismatch in the inspected window.

- The data-driven model is evaluated through its disturbed online Q_c proxy rather than through an explicit hard SOC reset parameter, which keeps the test aligned with deployment-time observables.

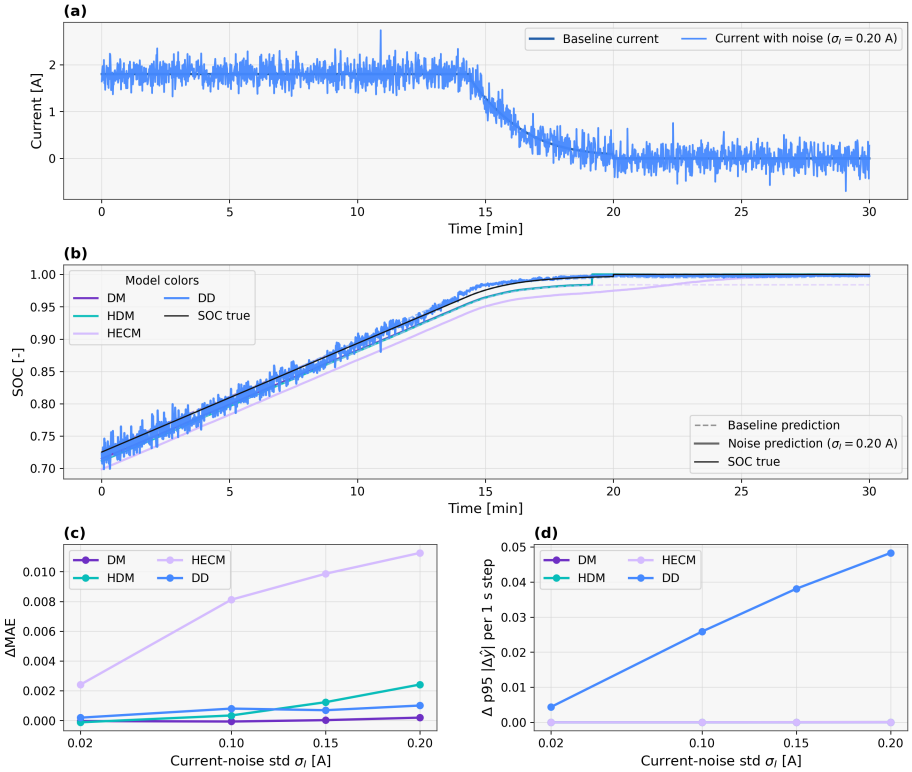

Figure 5: Noise robustness under additive current noise. The figure combines an applied current-noise example, a matched SOC comparison in the same 30 min window, the global MAE penalty, and the local output-jitter sensitivity across noise levels.

6.5. Signal integrity: missing samples, irregular sampling, and burst gaps

- Missing samples are the strongest global signal-integrity stressor in the benchmark, particularly for DM and HDM, whereas the data-driven model changes only marginally.

- The irregular-sampling scenario is nearly neutral for all models under the current resampled implementation, with only very small signed MAE changes around zero.

- The one-hour burst dropout remains globally moderate, but all models require roughly 27–29 h to return to the baseline error band, making the post-

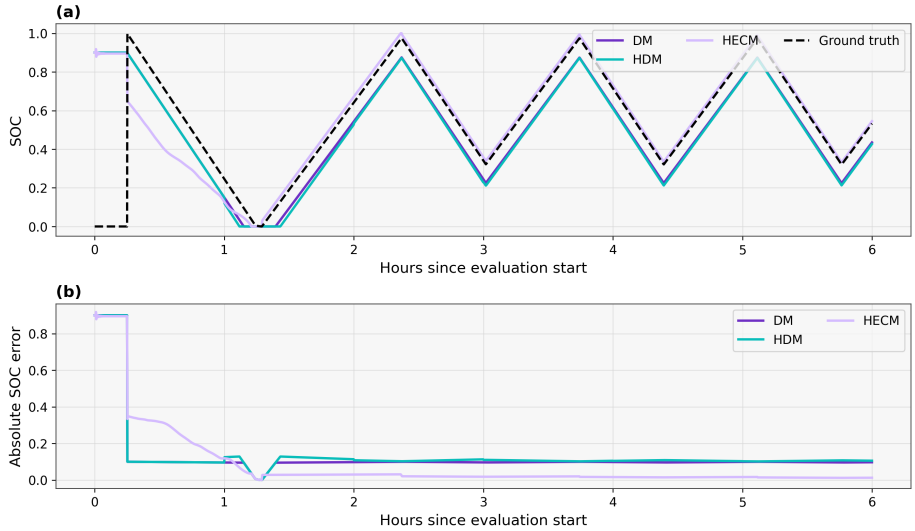

Figure 6: Initial-state recovery in a CV-free window. The top panel shows SOC trajectories after the imposed state mismatch, and the lower panel shows the corresponding absolute SOC error over time.

gap recovery behavior more informative than the global MAE alone.

- The signal-integrity block therefore needs both global and local views: Figure 7 summarizes the global penalties, while the two following figures show the transition into the gap and the long recovery afterwards.

6.6. Spike robustness

- Single voltage spikes have very different local effects across model classes, even when the full-run MAE changes remain small for DM, HDM, and HECM.
- The data-driven model shows the clearest spike sensitivity, both in the representative SOC window and in the local aligned error response.
- To avoid misleading signed full-run averages, the compact spike summary is reported as the 95th percentile of the post-spike peak excess error rather than as signed ΔMAE.

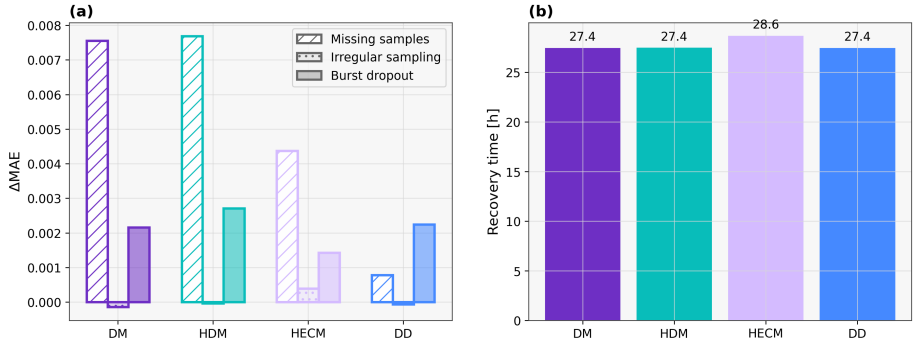

Figure 7: Signal-integrity robustness summary. The left panel compares the global MAE penalties for missing samples, irregular sampling, and burst dropout, whereas the right panel summarizes the burst-dropout recovery time.

6.7. Cross-scenario summary and ranking

- include a delta-MAE heatmap across scenarios and models.
- include a ranking table per scenario.
- identify which scenario is most discriminative for each estimator family.
- summarize where baseline performance and robustness rankings disagree.

7. Discussion

7.1. When each estimator fails and why

- discuss the direct-measurement model as the canonical drift-prone estimator under current bias and missing measurement information.
- discuss the hybrid direct-measurement model as a class with strong clean-condition calibration but no protection against integration drift.
- discuss the hybrid equivalent-circuit model as the strongest candidate under current bias because voltage residuals provide corrective information.
- discuss the data-driven model as a strong general-purpose estimator whose robustness depends on the type of corruption: good under missing samples, moderate under current bias, and more exposed to large voltage spikes.

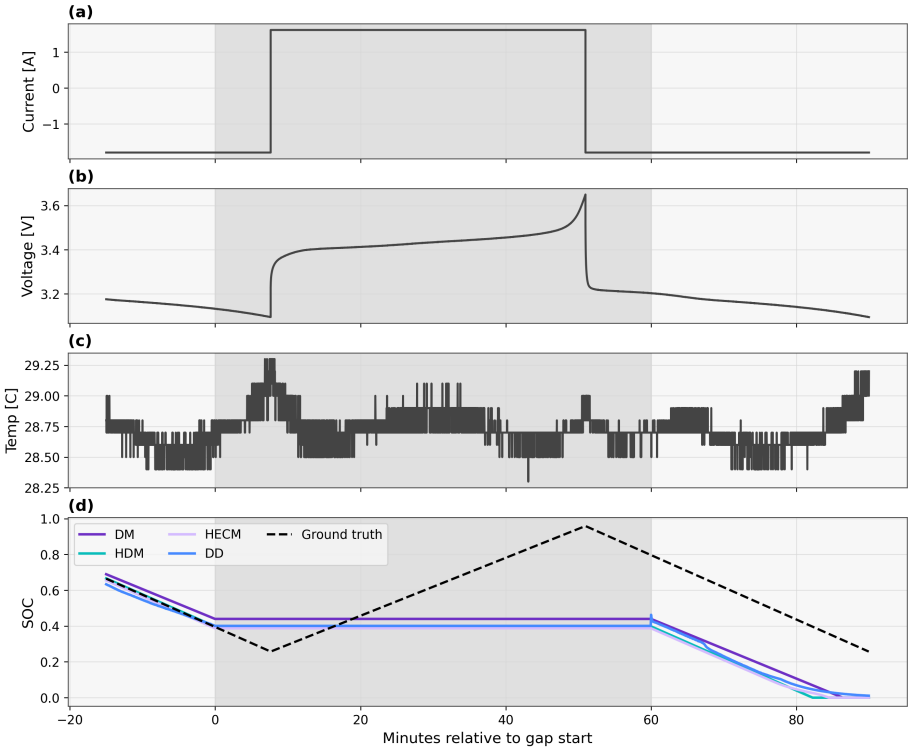

Figure 8: Local transition into the burst-dropout window. The shaded region marks the missing-data interval, and the lower panel shows how the estimators behave while the measurements are frozen.

7.2. Decision matrix for embedded BMS selection

- define one row per deployment condition, for example current sensor quality, voltage sensor quality, computational budget, and need for graceful degradation.
- define one column per estimator family.
- fill in preferred estimator choices based on robustness evidence instead of only baseline error.
- explain where a combined strategy could be useful, such as a data-driven estimator with explicit reset logic or a hybrid equivalent-circuit fallback.

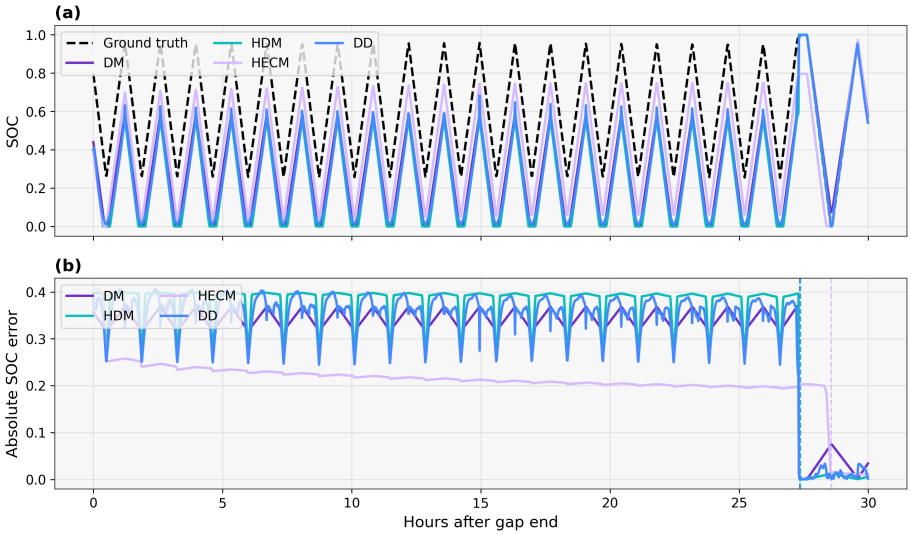

Figure 9: Post-gap recovery after the burst dropout. The upper panel shows the SOC trajectories after the gap, and the lower panel shows the corresponding absolute SOC error together with the recovery markers.

7.3. Limitations and threats to validity

- current benchmark phase is cell-specific and should later be expanded to cross-cell validation.
- the spike study currently uses isolated spikes; burst disturbances may expose stronger local differences.
- the initial-state test for the data-driven model uses a justified online-state proxy rather than a native explicit SOC-state parameter.
- extended scenarios such as voltage noise, temperature noise, offset, and quantization must still be integrated into the final result narrative once the campaign is complete.

8. Conclusion and Outlook

- summarize the paper as a robustness benchmark, not only an accuracy comparison.

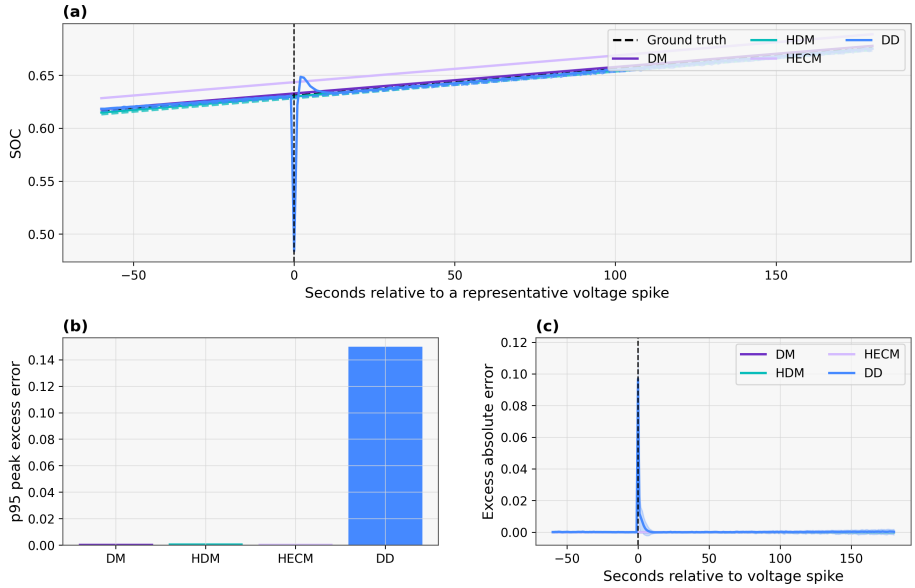

Figure 10: Voltage-spike robustness. The figure combines a representative SOC window around a spike, a local spike-severity summary based on the 95th percentile of the post-spike peak excess error, and the aligned local excess-error response across repeated spikes.

Placeholder for delta-MAE heatmap across models and scenarios.

`figures/Results/02_delta_mae_heatmap.png`

Figure 11: Planned cross-scenario heatmap showing error increase relative to baseline.

- state the main engineering message that estimator selection depends on expected sensor faults and recovery requirements.
- summarize the strongest confirmed findings already available: current-offset vulnerability of the direct-measurement family, bias robustness of the hybrid equivalent-circuit model, strong baseline of the hybrid direct-measurement model, and good missing-sample stability of the data-driven model.
- state that the benchmark is extensible and that the extended scenario block is already prepared in the simulation environment.
- define the next concrete expansion steps: full extended-matrix evaluation,

result-package finalization, and manuscript transition from bullet outline to continuous prose.

Data and Code Availability

- dataset citation and access note.
- simulation environment root and campaign scripts.
- note that the paper-facing results are collected under `DL\_Models/LFP\_SOC\_SOH\_Model/4\_simulation\_environment/results`.

References